

Physical equilibrium

Learning outcomes

By the end of this section you should be able to describe the characteristics of a physical system at equilibrium.

You will begin this section by trying a simple activity that illustrates the concept of physical equilibrium. The procedure is as follows:

1. Fill two beakers half-full with water. Mark the water level using a marker.
2. Cover one of the beakers with plastic wrap and leave the other one open.
3. Place both beakers in a sunny place and leave them for a few days.

Figure 1 shows what the beakers might look like after a few days.

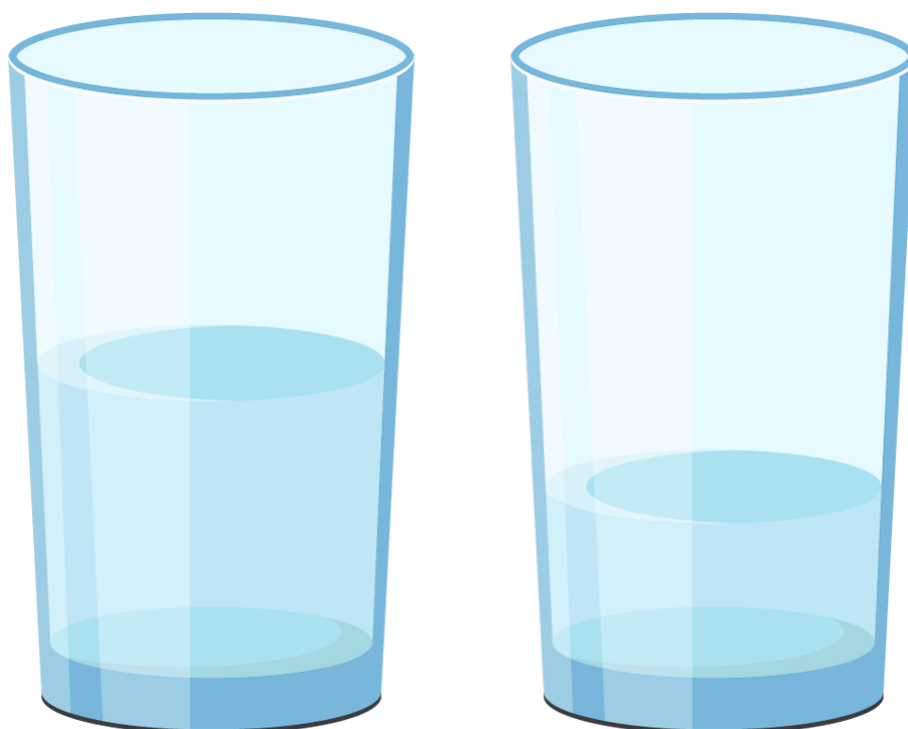
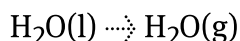


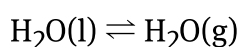
Figure 1. The beakers might look like this after a few days.

When you return to look at the beakers what do you notice? Has the water level changed in the beaker that was left open? What about in the beaker that was covered? How can you explain this? In the open beaker the water level has probably decreased quite considerably because the water evaporated. This can be represented as:



The state symbol of the water has changed from (l) liquid to (g) gas. Also notice the use of a one-way arrow ($\rightarrow$) which suggests that only the forward process is occurring.

Now consider what happened in the beaker that was covered with the plastic wrap. You may have noticed that the water level has decreased slightly but nowhere near as much as the open beaker. Inside the beaker the water is evaporating but the reverse process, condensation, is also taking place. This can be represented as:



Although it might look similar to the previous equation, there is one very important difference, which is the use of a double arrow ($\rightleftharpoons$). This double arrow means that the forward process (evaporation) and the reverse process (condensation) are both taking place at the same rate. Once these two rates are equal, the system has reached what is known as physical equilibrium (plural equilibria). Physical equilibrium involves a physical change and occurs when the rate of the forward process is equal to the rate of the reverse process. The equilibrium is often referred to as being 'dynamic' because both forward and reverse processes are still occurring.

A closed system is required for a system to reach equilibrium. A closed system is one where only energy can be exchanged between the system and the surroundings and no matter can escape. This explains why the open beaker did not reach equilibrium but the sealed beaker did because the water was able to escape from the open beaker (an open system) but not from the sealed beaker (a closed system). **Figure 2** summarises the differences between an open system and a closed system.

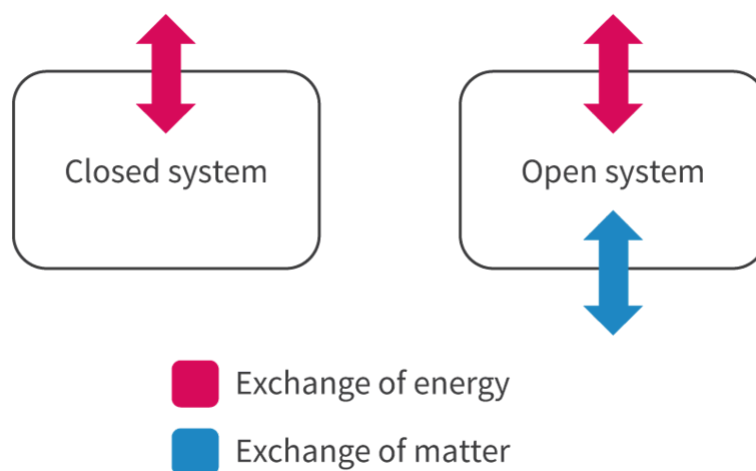
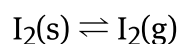


Figure 2. A closed system is required for a system to reach equilibrium.

📄 More information for figure 2

The next example illustrates another important point about a system at equilibrium. Iodine, when heated, will sublime to form iodine vapour (**Video 1**). The iodine vapour will undergo deposition back to form solid iodine. This can be represented as:



Once the system reaches equilibrium, you will notice that the purple colour inside the flask remains constant. Colour is a macroscopic property meaning that it can be seen with the unaided eye. So for a system at equilibrium, there is no change in macroscopic properties.

Watch **Video 1** and note the point at which the system reaches equilibrium.

Chemistry experiment 47 - Sublimation of Iodine



Video 1. At equilibrium, the purple colour inside the beaker remains constant. This is a visible indicator that the system has reached equilibrium.

👁 More information for video 1

Theory of Knowledge

Equilibrium in a chemical reaction can be seen as a competition between the formation of the products and formation of the reactants. Outside of chemistry, we also see opposing forces in many different contexts. In what situations does equilibrium exist outside the subject of science?

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

1. Write equations for the following physical equilibria.
 - (a) The equilibrium between solid water and liquid water.
 - (b) The equilibrium between liquid bromine and gaseous bromine.
 - (c) The equilibrium between aqueous carbon dioxide and gaseous carbon dioxide.
2. Describe the type of system required for a physical equilibrium to be established.
3. Explain why clothes that are hung on a washing line on a hot day dry quickly.
4. Explain why wet clothes sealed in a plastic bag do not dry.
5. Bromine is a reddish-brown liquid which forms a similarly coloured vapour. Outline how you would be able to tell if an equilibrium has been established in a sealed flask containing liquid bromine.

1. (a) $\text{H}_2\text{O(s)} \rightleftharpoons \text{H}_2\text{O(l)}$
(b) $\text{Br}_2\text{(l)} \rightleftharpoons \text{Br}_2\text{(g)}$
(c) $\text{CO}_2\text{(aq)} \rightleftharpoons \text{CO}_2\text{(g)}$
2. For a physical equilibrium to be established, a closed system is required. This is one in which only energy can be transferred between the system and the surrounding. Matter cannot escape which allows the equilibrium to be established.
3. Wet clothes dry quickly on a hot day because the rate of the evaporation is much greater than the rate of condensation. Effectively there is an open system which allows the matter (water) to escape which does not allow the equilibrium to be established.
4. Wet clothes sealed in a plastic bag do not dry because inside the bag, the rate of evaporation is equal to the rate of condensation. The sealed bag is a closed system which allows the equilibrium to be established.
5. Once the bromine starts to evaporate it would form a reddish-brown vapour. Once the reddish-brown colour (a macroscopic property) remains constant, you would know that an equilibrium has been established between the liquid bromine and the gaseous bromine.

R2.3 How far? The extent of chemical change

Chemical equilibrium

Learning outcomes

By the end of this section you should be able to describe the characteristics of a chemical system at equilibrium.

In section R2.3.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-equilibrium-id-43483/>), a physical process described as occurring in a closed system will reach an equilibrium position in which the rate of the forward process is equal to the rate of the reverse process. In this section, you will learn about chemical equilibrium which involves a chemical change rather than a physical change, but what is the difference between the two? Use your knowledge of physical and chemical changes to complete the drag and drop activity in **Interactive 1**.

Physical changes	Chemical changes
☐ Involve a physical change such as evaporation	☐ Involve a change in the chemical composition of the matter
☐ Involve the breaking of bonds between atoms	☐ No change in the chemical composition of the matter
☐ Involve a chemical reaction	☐ Involve the overcoming of intermolecular forces between atoms or molecules
☐ New products are formed	☐ No new products are formed

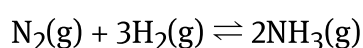
☒ Check

Interactive 1. Physical and Chemical Changes.

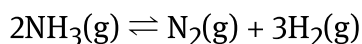
Keep these differences in mind as you study this section which focuses on chemical changes rather than physical changes.

A chemical system at equilibrium has much the same characteristics as a physical system at equilibrium but with one important difference - the system involves a chemical change rather than a physical change. One of the most important industrial chemical reactions is the Haber process (see [section R2.3.0](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)). In this reaction, hydrogen and nitrogen react together to form ammonia, NH_3 .

The forward reaction is:



and the reverse reaction is:



When nitrogen and hydrogen are mixed in a sealed reaction vessel, which is a closed system, the forward reaction proceeds as nitrogen and hydrogen react to produce ammonia. Once the concentration of ammonia starts to increase, the reverse reaction, in which ammonia decomposes to produce nitrogen and hydrogen, starts to take place. Eventually, the rate of the forward reaction will become equal to the rate of the reverse reaction. At this point, the system has reached equilibrium. The equilibrium is referred to as being dynamic because both the forward and reverse reactions are still taking place. This can be represented in graphical form as shown in **Figure 1**. Why do you think the reverse reaction starts from zero and then increases?

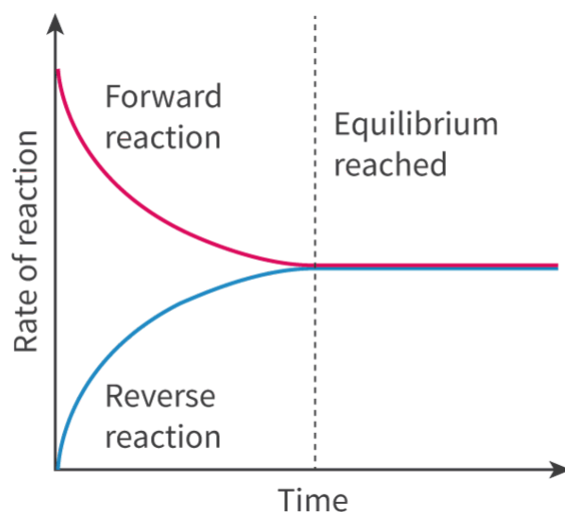


Figure 1. A chemical reaction reaches equilibrium when the rate of the forward reaction is equal to the rate of the reverse reaction.

🕒 More information for figure 1

Initially, the concentration of the product (ammonia) is zero, so there aren't any molecules to participate in the reverse reaction, which means the rate is also zero. As the concentration of the ammonia increases, the rate of the reverse reaction also increases.

The next characteristic of a chemical reaction at equilibrium involves the concentrations of reactants and products. The rate of reaction is defined as the change in concentration per unit time section R2.2.1

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-of-reaction-id-43481/>). How could the rate of reaction for the Haber process be followed in which all species are gases?

The rate of reaction could be monitored by the change in concentration of the reactants or of the products.

But what would this look like when both the forward and reverse reactions are occurring?

Figure 2 shows a graph of the change in the concentrations of nitrogen, hydrogen and ammonia as the forward reaction proceeds. The concentrations of nitrogen and hydrogen decrease as they react together and produce ammonia and the concentration of ammonia starts from zero and increases as initially the reaction vessel only contains nitrogen and hydrogen. What do you notice about the concentrations of the reactants and products once the reaction reaches equilibrium?

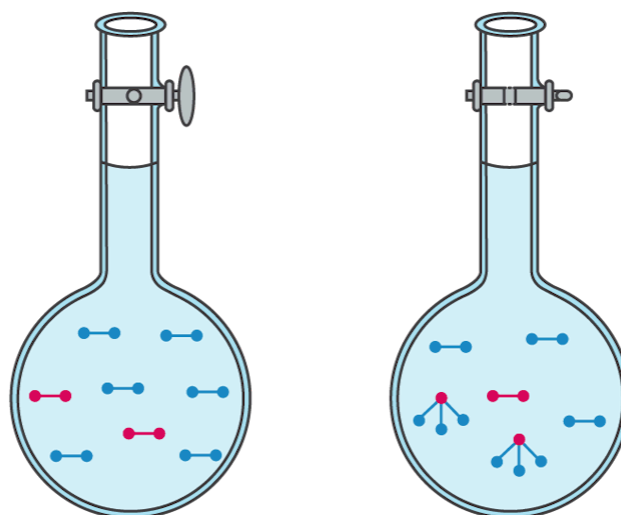
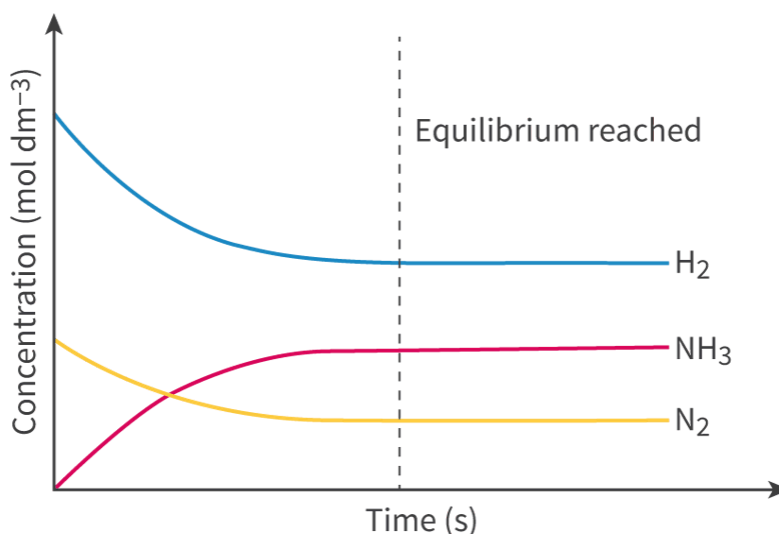


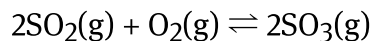
Figure 2. Once a reaction reaches equilibrium the concentrations of reactants and products remain constant.

🕒 More information for figure 2

You may have noticed that once the reaction reaches equilibrium, the concentrations of the reactants and products do not change, in other words they remain constant. This is an important point about a reaction at equilibrium: the rates of the forward and reverse reactions are equal but the concentrations of reactants and products are not. It is also important to note that at equilibrium, the concentration of reactants and products are rarely equal. It is much more common to have higher concentrations of reactants or products when a reaction reaches equilibrium.

A chemical reaction can reach equilibrium from either direction starting with the reactants or with the products. **Figure 3** and **Figure 4** show this for the reaction between sulfur dioxide and oxygen to produce sulfur trioxide and also for the reverse reaction in which sulfur trioxide reacts to form sulfur dioxide and oxygen.

The equation for the forward reaction is:



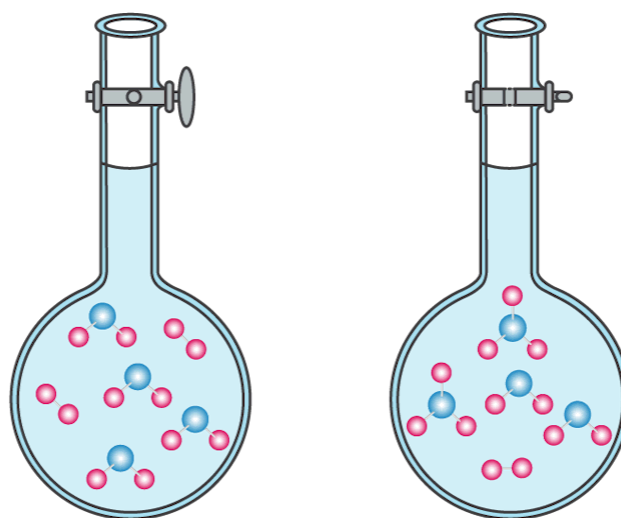
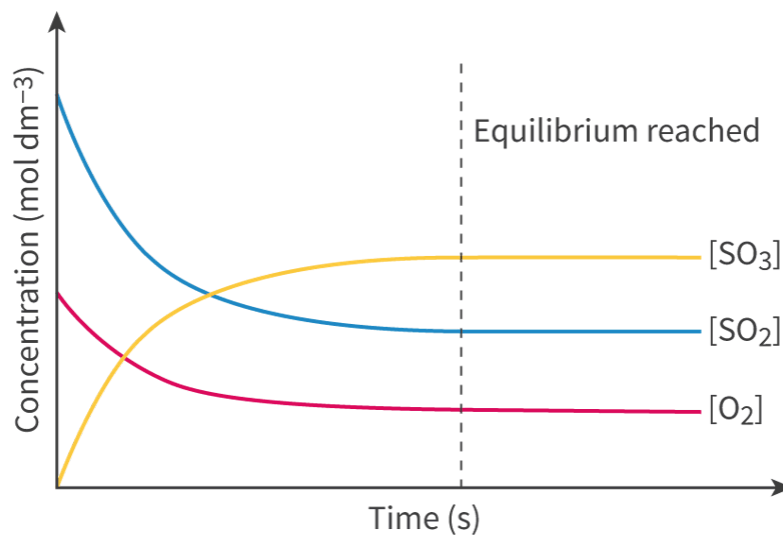


Figure 3. The equilibrium position can be reached from the forward reaction or the reverse reaction.

© More information for figure 3

And for the reverse reaction:



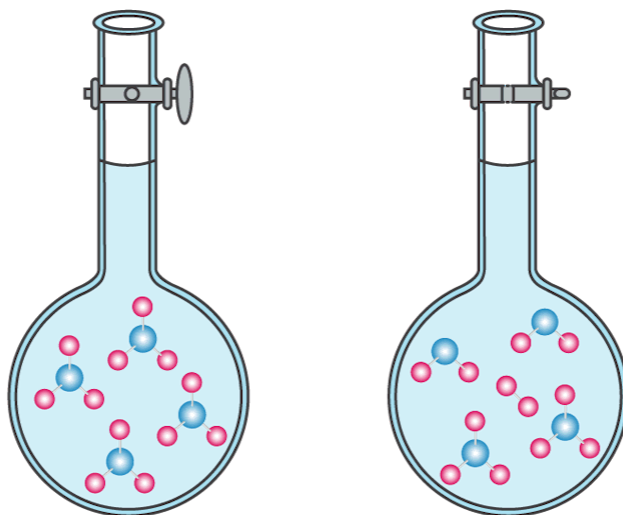
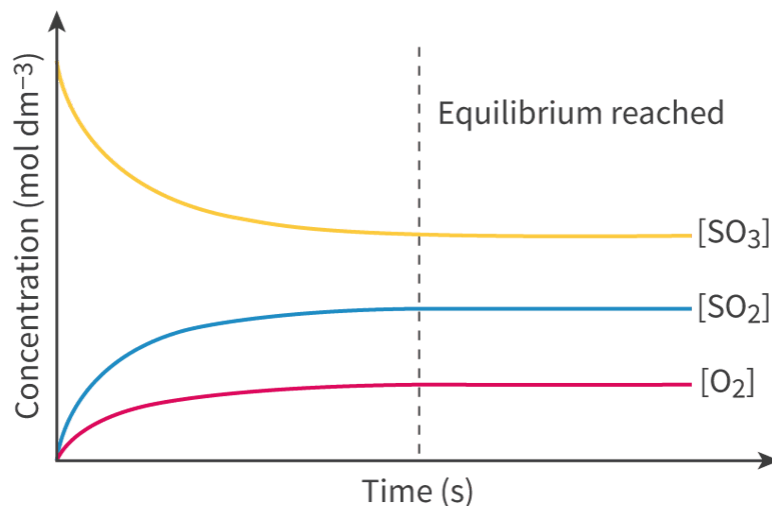


Figure 4. The equilibrium position can be reached from the forward reaction or the reverse reaction.

© More information for figure 4

From the graphs, you can see that from whichever side the reaction starts (reactants or products), the equilibrium position is reached when the concentration of reactants and products remain constant.

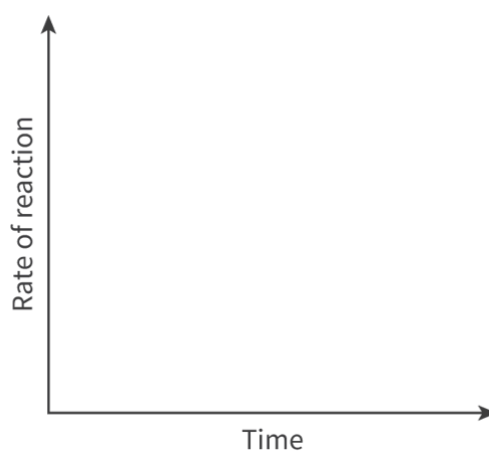
Study skills

You might have noticed that all the examples of equilibrium covered so far involve reaction where the reactants and products are in the same state. These are known as homogeneous reactions or examples of homogeneous equilibria. There are also equilibrium systems where the reactants and products are in different states, known as heterogeneous equilibrium, such as solid ice cubes and liquid water.

Activity

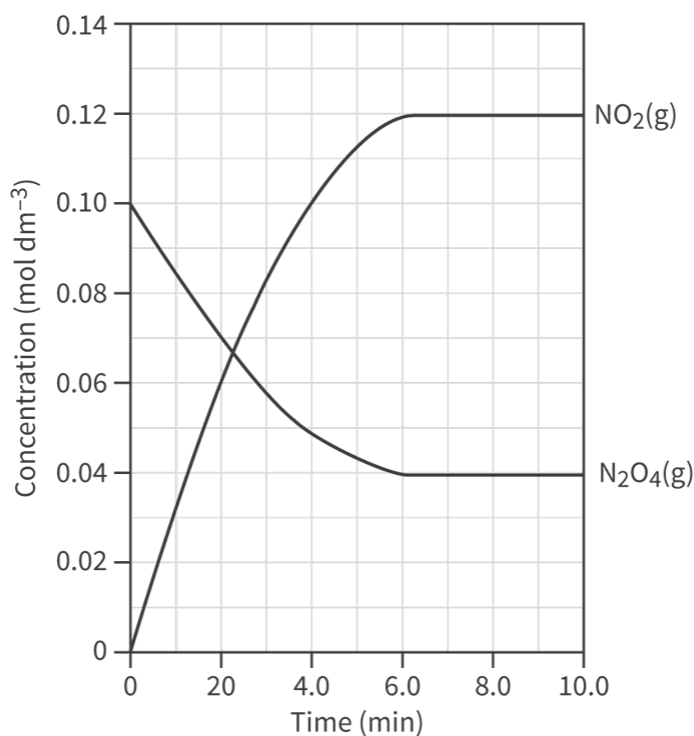
- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

1. When a chemical reaction is at equilibrium, state how the rate of the forward reaction compares to the rate of the reverse reaction.
2. Outline the difference between using a one-way arrow ($\rightarrow$) and a double arrow ($\rightleftharpoons$) for a chemical reaction.
3. Sketch a graph on the axis below to represent the forward and reverse reactions for a reversible reaction. Label the point at which the reaction reaches equilibrium.



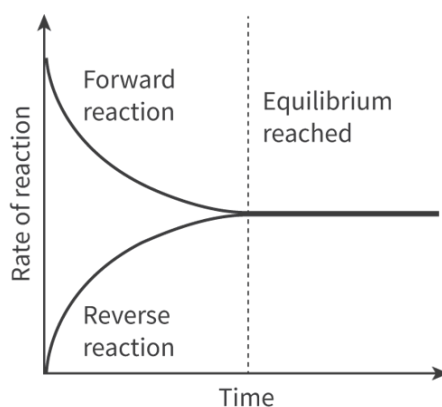
 More information

4. The graph below shows how the concentrations of $\text{NO}_2(\text{g})$ and $\text{N}_2\text{O}_4(\text{g})$ change throughout the course of a reaction. Determine the time at which the reaction reaches equilibrium giving a reason for your answer.



👁 More information

1. For a reaction at equilibrium, the rate of the forward reaction is equal to the rate of the reverse reaction.
2. A one-way arrow implies that the reaction proceeds in the forward direction only. A double arrow implies that the reaction is reversible meaning that both the forward and reverse reactions occur.
3. The graph should look like the one shown below. The position of equilibrium is reached once the rate of the forward reaction is equal to the rate of the reverse reaction.



👁

4. The reaction reaches equilibrium at six minutes. The concentrations of reactants and products remain constant at equilibrium.

The equilibrium constant, K_c and magnitude of the K_c

Learning outcomes

By the end of this section you should be able to:

- (R2.3.2) Deduce the equilibrium constant equation from an equation for a homogeneous reaction.
- (R2.3.3) Relate the magnitude of the equilibrium constant for a reaction to the composition of the reaction mixture at equilibrium.
- (R2.3.3) Determine the relationships between the equilibrium constants for reactions that are the reverse of each other at the same temperature.

In the Big Picture (see [section R2.3.0 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/)), the concept of ‘extent of reaction’ was introduced. This can be thought of as how far to the right (the products side) a reaction proceeds. Different reactions proceed to different extents which means that some reactions proceed further to the right than others. But how do we know how far a reaction proceeds? Is there a way to quantify the concept of the extent of a reaction? The answer is yes and in this section you will learn about how the magnitude of the equilibrium constant, K_c , can be used as an indicator of the extent of a chemical reaction.

To begin this section, you will learn how to write the equilibrium constant expression for a reaction at equilibrium.

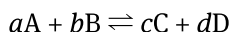
Study skills

The symbol for the equilibrium constant is usually written as K_c but can also be written as K . Be careful to distinguish this from the symbol used for the temperature in kelvin which has the symbol K but is not written in italics.

The equilibrium constant, K_c , of a reaction is a ratio that provides information about the composition of a reaction mixture at equilibrium. In other words, it gives information about the extent of the reaction. The concentrations of reactants and products in the equilibrium mixture are determined from experimental data.

To calculate the value of the equilibrium constant for a reaction, we need to first write an equilibrium constant expression for the reaction.

To do so, let's consider the general reaction between reactants A and B to form products C and D, where a , b , c and d are coefficients.



The equilibrium constant expression for this reaction is written as:

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

- Square brackets are used to represent the concentrations of the reactants and products in mol dm^{-3}
- The products are placed in the numerator and the reactants in the denominator.
- The concentrations of the reactants and products are raised to the power of their coefficients in the balanced equation.

State symbols are usually left out of the expression and the equilibrium constant is dimensionless (it has no units). It should also be noted that you are only required to write equilibrium constant expressions for homogeneous reactions, where all reactants and products are in the same state. This usually involves reactions where all species are present in the gaseous state (g), but can also include reactions where all species are present in the liquid state (l).

Let's look at the equilibrium constant expression for the Haber process in which nitrogen gas and hydrogen gas react to form ammonia.

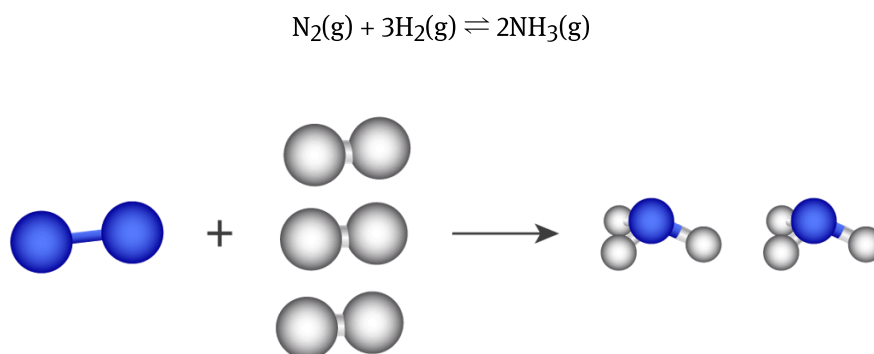


Figure 1. A molecular representation of the Haber process.

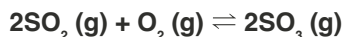
© More information for figure 1

The equilibrium constant expression for the reaction is:

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2] [\text{H}_2]^3}$$

Use **Interactive 1** to test your understanding of writing an equilibrium constant expression. Drag and drop the parts to the correct places in the expression.

Arrange the reactants and products into the correct places to construct the equilibrium constant equation for the reaction:



$$K_c = \underline{\hspace{2cm}}$$

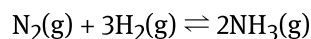


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Interactive 1. Writing an equilibrium constant expression.

The magnitude of the equilibrium constant

If the equilibrium concentrations of reactants and products are known, the value of the equilibrium constant can be calculated. Take for example, the reaction between nitrogen and hydrogen to produce ammonia.



The equilibrium constant expression for the reaction is:

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2] [\text{H}_2]^3}$$

Table 1. The equilibrium concentrations of the reactants and products at a temperature of 500 °C

	Reactants		Products
	N ₂ (g)	H ₂ (g)	NH ₃ (g)
Equilibrium concentration (mol dm ⁻³)	0.80	2.40	0.40

Using this data, the equilibrium constant for the reaction can be calculated. To do this the equilibrium concentrations of the reactants and products are substituted into the equilibrium constant expression for the reaction:

$$\begin{aligned} K_c &= \frac{[0.40]^2}{[0.80][2.40]^3} \\ &= 0.01446\dots \\ &= 0.014 \text{ (2 s.f.)} \end{aligned}$$

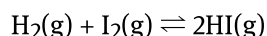
Now that we have calculated the equilibrium constant for the reaction at this temperature, what does this value tell us? It is relatively small, so do you think that there is a higher concentration of reactants or products at equilibrium? What if the value of the equilibrium constant is very large? The value of the equilibrium constant for a reaction tells us the extent of a reaction, or more specifically, to which side the equilibrium position lies. The equilibrium position can either lie to the left (reactants side) or to the right (products side). If the equilibrium position lies to the left, there is a greater concentration of reactants present in the equilibrium mixture. Conversely, if the equilibrium position lies to the right, there is a greater concentration of products present in the equilibrium mixture. This is summarised in **Table 2**.

Table 2. Equilibrium position.

Equilibrium position	Composition of equilibrium mixture
Lies to the right	Higher concentration of products
Lies to the left	Higher concentration of reactants

For the earlier example involving the Haber process, the value of the equilibrium constant was relatively small (0.0145) which means that at equilibrium, there is a higher concentration of reactants than products in the equilibrium mixture. So, for this reaction at this temperature, we say that the equilibrium position lies to the left.

In the next example, we have the reaction between hydrogen and iodine to produce hydrogen iodide.



Using the values in **Table 3**, calculate the value of the equilibrium constant.

Table 3. Equilibrium concentrations for the reaction of hydrogen and iodine.

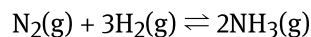
	Reactants		Products
	$H_2(g)$	$I_2(g)$	$HI(g)$
Equilibrium concentration ($mol\ dm^{-3}$)	0.0037	0.0037	0.028

$$\begin{aligned}
 K_c &= \frac{[HI]^2}{[H_2][I_2]} \\
 &= \frac{[0.028]^2}{[0.0037][0.0037]} \\
 &= 57.268\dots \\
 &= 57 \text{ (2 s.f.)}
 \end{aligned}$$

What does this value of the equilibrium constant tell us about the composition of the reaction mixture at equilibrium? Compared to the previous example, this value of the equilibrium constant is considerably larger. So for this reaction at this temperature, there is a higher concentration of products than reactants. Therefore, we say that the equilibrium position lies to the right.

Worked example 1

At equilibrium, the reaction:



contains:

$$\begin{aligned}
 &0.30\ mol\ dm^{-3}\ H_2(g) \\
 &0.10\ mol\ dm^{-3}\ N_2(g) \\
 &0.80\ mol\ dm^{-3}\ NH_3(g)
 \end{aligned}$$

Determine the value of the equilibrium constant and state which side the equilibrium position lies.

$$\begin{aligned}
 K_c &= \frac{[NH_3]^2}{[N_2][H_2]^3} \\
 &= \frac{[0.80]^2}{[0.10][0.30]^3} \\
 &= 237.037\dots \\
 &= 240 \text{ (2 s.f.)}
 \end{aligned}$$

The equilibrium constant is much greater than one, so the equilibrium position lies to the right and the mixture contains mostly products.

Figure 2 shows how the value of the equilibrium constant can be used to determine the composition of the equilibrium mixture for a reaction at equilibrium.

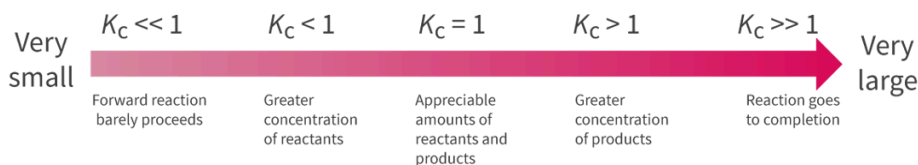


Figure 2. The value of the equilibrium constant is used to determine the composition of the equilibrium mixture.

📄 More information for figure 2

Table 4. A summary of equilibrium constant values and the composition of the reaction mixture

Value of K_c	Composition of reaction mixture
$K_c \ll 1$	Almost exclusively reactants
$K_c < 1$	Higher concentration of reactants
$K_c = 1$	Appreciable amounts of reactants and products
$K_c > 1$	Higher concentration of products
$K_c \gg 1$	Almost exclusively products

Making connections

How does the value of K for the dissociation of an acid convey information about its strength? [subtopic R3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/)

Now try the drag and drop activity in **Interactive 2** to test your understanding of equilibrium constant values and the composition of the reaction mixture.

K_c value	Composition of the reaction mixture	Equilibrium position
4.61×10^{-3}		
640		
1.9×10^{-23}		
3.2×10^{81}		

Lies to the left

Forward reaction barely proceeds

Lies to the right

Almost exclusively products

Almost exclusively reactants

Higher concentration of products

Reaction goes to completion

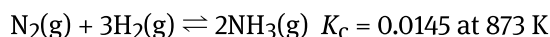
Higher concentration of reactants

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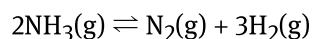
Interactive 2. Equilibrium constant values and the composition of the reaction mixture.

Determine the equilibrium constant for the reverse reaction at the same temperature

Earlier in the section, we looked at the equation for the Haber process and its equilibrium constant at 873 K.



But what about the equilibrium constant for reverse reaction at the same temperature?



The equilibrium constant equation for the reverse reaction is:

$$K_c = \frac{[\text{N}_2][\text{H}_2]^3}{[\text{NH}_3]^2}$$

Next, input the equilibrium concentrations of reactants and products into the

$$K_c = \frac{[0.80][2.40]^3}{[0.40]^2}$$

equation and solve for K_c .

$$= 69.12$$

$$= 69 \text{ (2 s.f.)}$$

Note that this is the reciprocal of the equilibrium constant for the forward reaction.

$$K_{c \text{ reverse}} = \frac{1}{K_{c \text{ forward}}}$$

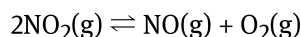
$$= \frac{1}{0.01446 \dots}$$

$$= 69.12$$

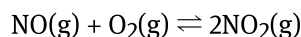
$$= 69 \text{ (2 s.f.)}$$

Worked example 2

The equilibrium constant for the reaction shown is 1.22×10^6 at a specific temperature.



Determine the equilibrium constant for the reaction below at the same temperature.



The reaction is the reverse of the original, so we can take the reciprocal of the equilibrium constant for the forward reaction.

$$K_{c \text{ reverse}} = \frac{1}{K_{c \text{ forward}}}$$

$$= \frac{1}{1.22 \times 10^6}$$

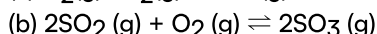
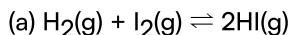
$$= 8.196 \dots \times 10^{-7}$$

$$= 8.20 \times 10^{-7} \text{ (3 s.f.)}$$

Activity

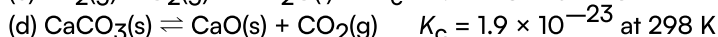
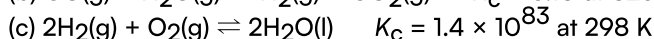
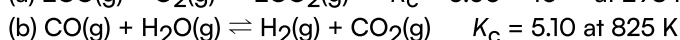
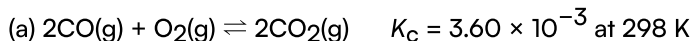
- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. Deduce the equilibrium constant equation for these reactions.



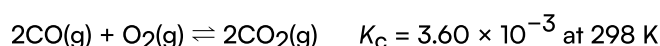
2. Outline what is meant by the term homogeneous reaction.

3. For each of the reactions below, determine the equilibrium position and the composition of the reaction mixture.

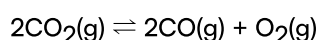


4. Explain the relationship between the magnitude of the equilibrium constant, the equilibrium position and the composition of the equilibrium mixture.

5. Carbon monoxide reacts with oxygen to produce carbon dioxide as shown.



Deduce the equilibrium constant for the following reaction at the same temperature.



1. (a) $K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$

(b) $K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]}$

2. The term homogeneous reaction refers to a reaction in which all the reactants and products are in the same state. It is also known as homogeneous equilibria.

3. (a) The equilibrium constant for this reaction is less than one ($K_c < 1$) so the equilibrium position lies to the left and the equilibrium mixture will contain a higher concentration of reactants.
 (b) The equilibrium constant for this reaction is greater than one ($K_c > 1$) so the equilibrium position lies to the right and the equilibrium mixture will contain a higher concentration of products.
 (c) The equilibrium constant for this reaction is much greater than one ($K_c \gg 1$) so the reaction goes to completion and the equilibrium mixture will consist of almost exclusively products.
 (d) The equilibrium constant for this reaction is much less than one ($K_c \ll 1$) so the forward reaction barely proceeds and the equilibrium mixture will consist of almost exclusively reactants.

4. The greater the value of the equilibrium constant, the further to the right the equilibrium position lies and the higher the concentration of products in the equilibrium mixture.

The smaller the value of the equilibrium constant, the further to the left the equilibrium position lies and the higher the concentration of reactants in the equilibrium mixture.

5. The equilibrium constant for the reverse reaction is the reciprocal of the equilibrium constant of the forward reaction.

$$\begin{aligned}K_c &= \frac{1}{3.60 \times 10^{-3}} \\&= 277.777\dots \\&= 278 \text{ (3 s.f.)}\end{aligned}$$

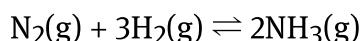
R2.3 How far? The extent of chemical change

Le Châtelier's principle

Learning outcomes

By the end of this section you should be able to apply Le Châtelier's principle to predict and explain responses to changes of systems at equilibrium.

The Haber process was first introduced in the Big Picture (see [section R2.3.0 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/)). The process is often described as 'making bread from air' because the nitrogen required for the reaction is obtained from the air and the ammonia produced is used to make fertiliser which in turn is used to grow crops for food production. The reaction itself is between nitrogen and hydrogen to produce ammonia.



During the Haber process, the conditions are carefully controlled to produce as much ammonia as possible, as economically as possible. These conditions include a very high pressure of approximately 20 000 kPa and a temperature of 450 °C. But why are such conditions necessary? The obvious answer would be to maximise the yield of ammonia. In this section, we see how the extent of a reaction can be influenced in order to maximise the yield of a chemical reaction.

Le Châtelier's principle is used to predict the effect on the equilibrium position when changes are made to a system at equilibrium. The equilibrium position can shift to the right or the left depending on the changes made to the reaction at equilibrium. If the equilibrium position shifts to the left, the concentration of reactants increases. Conversely if the equilibrium position shifts to the right, the concentration of products increases.

Le Châtelier's principle states that:

'When a system at equilibrium is subjected to a change, the system will respond to minimise the effect of the change.'

In its simplest form, it states that if changes are made to a system at equilibrium, the system will try to counteract the changes made. For example, if the temperature of a system at equilibrium is increased the system will try to decrease the temperature. Or if the pressure of a system at equilibrium is increased, the system will try to decrease the pressure.

The changes that can be made to a system at equilibrium are as follows:

1. Changes made to the concentration by either removing a reactant or product from the system at equilibrium.
2. Changes made to the pressure of a gaseous reaction by changing the volume of the reaction vessel, in accordance with Boyle's law (see [section S1.5.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ideal-gas-equation-id-44353/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ideal-gas-equation-id-44353/>)).
3. Changes made to the temperature of the system at equilibrium by either adding or removing heat.

Changes in concentration

First, we will consider changes in concentration. Again, we will use the reaction for the Haber process to explain the effect that changing the concentration has on the equilibrium position. **Figure 1** shows a graph of the concentrations of nitrogen, hydrogen and ammonia during the course of the reaction. The reaction is at equilibrium from 0 to 5 minutes as shown by the constant concentrations of the reactants and products. At five minutes, nitrogen is added to the reaction vessel which is shown on the graph by the vertical line. What effect does this have on the concentrations of hydrogen and ammonia? Study the graph and check your answer below when you are ready.

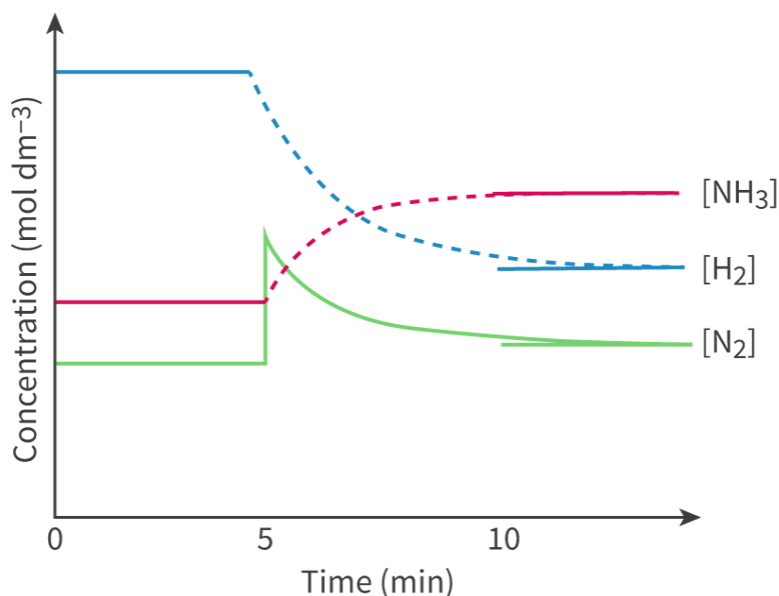


Figure 1. The effect of adding nitrogen on the concentrations of the reactants and products of the equilibrium mixture.

 More information for figure 1

This graph shows the concentrations of ammonia (NH_3), hydrogen (H_2), and nitrogen (N_2) over time during the Haber process reaction. The X-axis represents time in minutes ranging from 0 to 15 minutes, while the Y-axis represents concentration in mol dm^{-3} . Nitrogen concentration is indicated by a green line, which sharply increases at 5 minutes where a vertical line is shown. At this point, the hydrogen (H_2) concentration, shown by a blue dashed line, begins to gradually decrease, while the ammonia (NH_3) concentration, indicated by a pink dashed line, shows a slight constant increase after the 5-minute mark. The graph illustrates how adding nitrogen affects the equilibrium of this chemical reaction.

[Generated by AI]

As shown on the graph, the concentration of hydrogen decreases and the concentration of ammonia increases.

What has caused these changes in the concentrations of reactants and products? According to Le Châtelier's principle, when the nitrogen was added, the system responded by using up the additional nitrogen to produce more ammonia. This explains the increase in the concentration of ammonia that is observed. The concentration of the hydrogen decreased because it reacted with the added

nitrogen to produce ammonia. To summarise, adding nitrogen to the system at equilibrium shifted the equilibrium position to the right (the products side) which resulted in an increase in the yield of ammonia.

Removing a product will cause the equilibrium position to shift to the right, increasing the yield of the products(s).

The removal of a reactant will shift the equilibrium position to the left, increasing the yield of the reactant(s).

Changes in either the concentration of a reactant or a product have no effect on the value of the equilibrium constant.

Changes in pressure

Changes to the pressure of a system at equilibrium apply to gaseous reactions only. The pressure of a system can be changed by changing the volume of the reaction vessel according to Boyle's law (**Figure 2**). A decrease in the volume of the reaction vessel will increase the pressure whereas an increase in the volume will decrease the pressure.

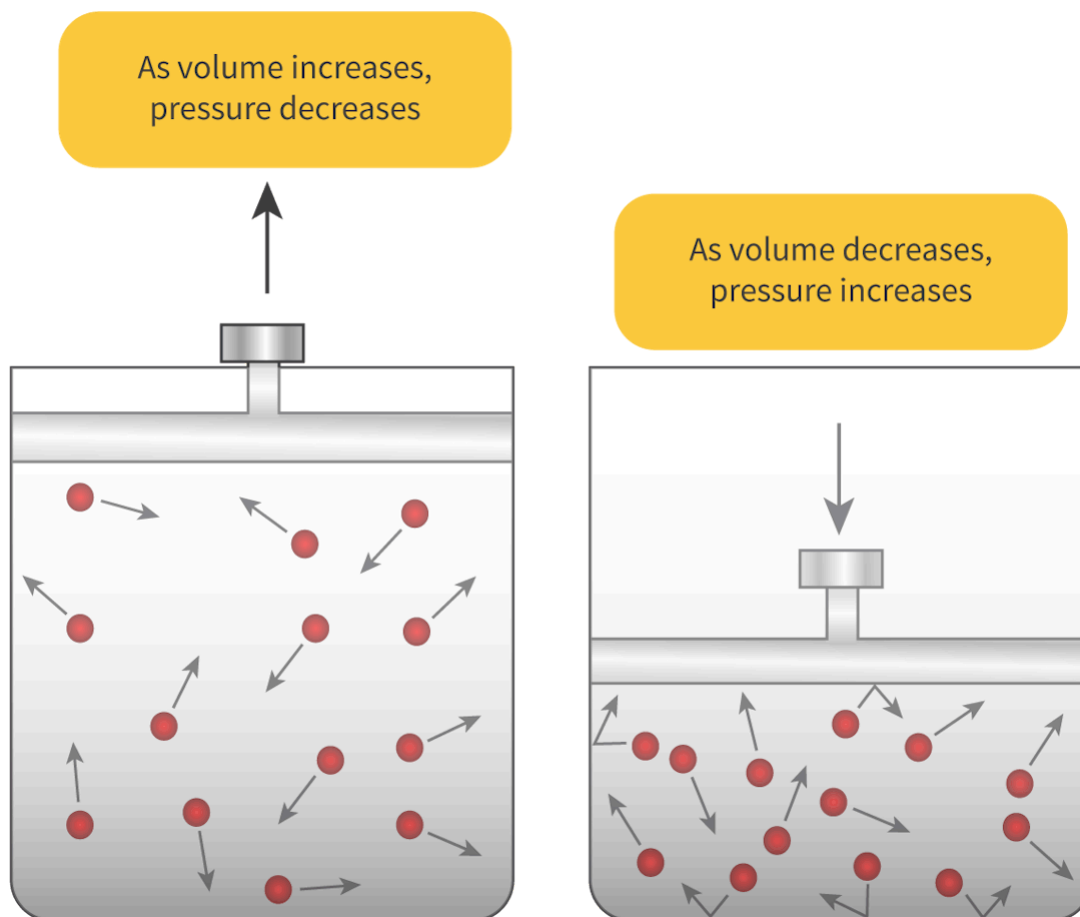


Figure 2. Boyle's law states that pressure and volume are inversely proportional at constant temperature.

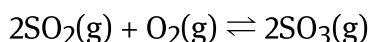
 More information for figure 2

The image depicts Boyle's law with two separate diagrams of a piston. On the left, the piston is raised, allowing for an increase in volume. Red spheres representing gas molecules are spread out with arrows indicating their movement. A yellow box above the diagram reads, 'As volume increases, pressure decreases,' illustrating the inverse relationship between volume and pressure at constant temperature.

On the right, the piston is lowered, resulting in a decrease in volume. The gas molecules are closer together, and their movement is denoted by arrows. The corresponding yellow box states, 'As volume decreases, pressure increases,' reinforcing Boyle's law principle. The illustration uses visual positioning and arrows to communicate the change in molecular movement corresponding to changes in volume and pressure.

[Generated by AI]

To determine which side the equilibrium position will shift to when the pressure of the system is changed, count the number of moles of gas on each side of the equation. This can be done by simply adding up the coefficients of the reactants and products in the balanced equation. For example, in the reaction between sulfur dioxide and oxygen to produce sulfur trioxide, we can see that there are a total of three moles of gas in the reactants and two moles of gas in the products.



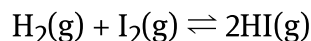
Once the number of moles of gas in the reactants and products have been determined, the shift in the equilibrium position can be predicted as follows:

- An increase in pressure favours the side with the fewer number of gaseous molecules. This occurs because at high pressure, gas particles will experience a greater frequency of collisions, increasing their rate of reaction.
- A decrease in pressure favours the side with the greater number of gaseous molecules. This occurs because at low pressure, the

particles are farther apart and have a lower frequency of collision, therefore a decrease in the rate of reaction.

So for the reaction shown above, what is the effect of changing the pressure of the system at equilibrium? If the pressure is increased, the equilibrium position shifts to the right because the product side has a fewer number of gaseous molecules. This has the effect of decreasing the pressure of the system. And conversely, if the pressure is decreased, the equilibrium position shifts to the left because the reactants side has a greater number of gaseous molecules. This has the effect of increasing the pressure of the system.

It is important to note that changes in pressure only have an effect on the equilibrium position when there are different numbers of gaseous molecules in the reactants and products. Take for example, the reaction between hydrogen and iodine to produce hydrogen iodide.



If the pressure of this system changes, there will be no effect on the equilibrium position because there are two moles of gas on each side of the equation.

Changes in pressure have no effect on the value of the equilibrium constant because the equilibrium constant only varies with a change in temperature.

Changes in temperature

To predict the effect of a change in temperature on the equilibrium position we need to know the enthalpy change, ΔH , for the forward (or reverse) reaction. A negative enthalpy change means that the forward reaction is exothermic [section R1.1.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/endothemic-and-exothermic-reactions-id-45282/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/endothemic-and-exothermic-reactions-id-45282/) with heat released from the system to the surroundings and the reverse reaction is endothermic with heat being absorbed from the surroundings to the system (**Figure 3**).

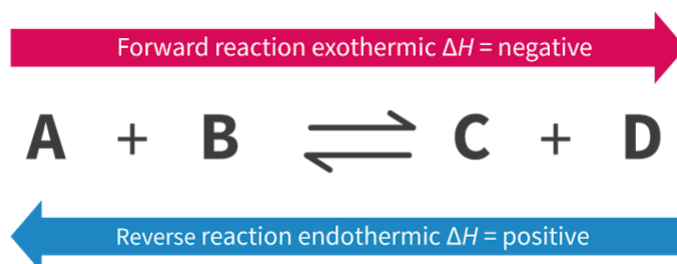


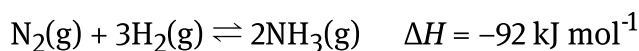
Figure 3. The enthalpy change of the forward reaction is the opposite of the reverse reaction.

 More information for figure 3

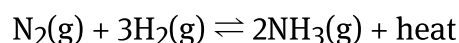
The diagram illustrates two opposing chemical reactions with their respective enthalpy changes. At the center, two reactants, labeled 'A' and 'B', are shown reacting to form two products, 'C' and 'D'. There is a double-headed arrow between the reactants and products, indicating the reversible nature of the reaction. Above this reaction, a pink arrow points to the right, labeled 'Forward reaction exothermic $\Delta H = \text{negative}$ ', indicating that the forward reaction releases heat to the surroundings and has a negative enthalpy change. Below the reaction, a blue arrow points to the left with the label 'Reverse reaction endothermic $\Delta H = \text{positive}$ ', signifying that the reverse reaction absorbs heat from the surroundings and has a positive enthalpy change.

[Generated by AI]

Returning to the reaction for the Haber process, we see it has a negative enthalpy change for the forward reaction.

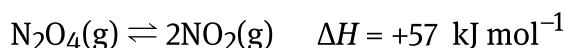


In this context, it is helpful to consider heat as either a reactant for an endothermic reaction or as a product for an exothermic reaction. So we can rewrite the equation for the Haber process as:



If heat is added to the system at equilibrium it can be thought of as the addition of a product which shifts the equilibrium position to the left in the endothermic direction and if heat is removed from the system at equilibrium, the equilibrium position will shift to right in the exothermic direction.

Next we will consider the effect of adding or removing heat on an endothermic reaction.



This equation can be rewritten to include the heat as a reactant:



Now we can see that adding heat to the system at equilibrium will shift the equilibrium position to the right in the endothermic direction and removing heat will shift the equilibrium position to the left in the exothermic direction.

Temperature and the equilibrium constant

Now we will look at the effect of changing the temperature on the value of the equilibrium constant. Recall that the equilibrium constant is temperature dependent meaning that its value changes if the temperature of the reaction is changed. If the equilibrium position shifts to the right, the value of the equilibrium constant will increase and if the equilibrium position shifts to the left, the value of the equilibrium constant decreases. So for an exothermic reaction, increasing the temperature shifts the equilibrium position to the left and the value of the equilibrium constant decreases and for an endothermic reaction, increasing the temperature shifts the equilibrium position to the right and the value of the equilibrium constant increases.

Try **Interactive 1** to test your understanding of temperature and equilibrium.

	Exothermic reaction	Endothermic reaction
Effect on equilibrium position when temperature increases		
Effect on equilibrium position when temperature decreases		
Effect on K_c when temperature increases		
Effect on K_c when temperature decreases		

Shifts to the left

Shifts to the right

 K_c decreases K_c increases

✓ Check

Interactive 1. Temperature and equilibrium.

Catalysts and equilibrium

You learned previously in [section R2.2.4–5](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>) that a catalyst increases the rate of a chemical reaction by providing an alternative reaction pathway with a lower activation energy. A catalyst added to a reversible reaction will allow the reaction to reach an equilibrium position sooner but has no effect on the equilibrium position or on the value of the equilibrium constant. This is because the rate of the forward reaction and the rate of the reverse reactions are increased by the same amount.

Making connections

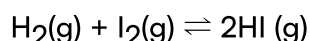
Why do catalysts have no effect on the value of the equilibrium constant or on the equilibrium composition subtopic R2.2

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43127/>)?

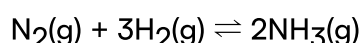
Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

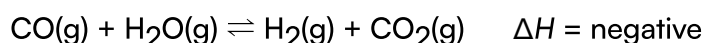
1. Predict the effect on the yield of hydrogen iodide and on the value of the equilibrium constant when the concentration of hydrogen is increased in the following reaction.



2. Predict the effect on the equilibrium position, the yield of ammonia and on the value of the equilibrium constant when the volume of the reaction vessel is decreased at constant temperature for the following reaction.



3. Predict the effect on the equilibrium position, the yield of hydrogen and on the value of the equilibrium constant when the temperature of the system is increased for the following reaction.



4. Explain the effect of adding a catalyst to a reaction at equilibrium.

1. Adding hydrogen will shift the equilibrium position to the right which will increase the yield of the hydrogen iodide. The value of the equilibrium constant will remain unchanged.
2. Decreasing the volume of the reaction vessel at constant temperature will increase the pressure of the system. The equilibrium position will shift to the side that has the fewer number

of gaseous molecules, which is the products side. The yield of ammonia will increase. The value of the equilibrium constant will remain unchanged.

3. Increasing the temperature for an exothermic reaction will shift the equilibrium position to the left in the endothermic direction. The yield of hydrogen will decrease as will the value of the equilibrium constant.
4. A catalyst has no effect on the equilibrium position because it increases the rate of the forward and reverse reactions by the same amount.

The reaction quotient, Q (HL)

Higher level (HL)

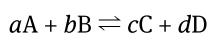
Learning outcomes

By the end of this section you should be able to calculate the reaction quotient, Q , from the concentrations of reactants and products at a particular time and determine the direction in which the reaction will proceed in order to reach equilibrium.

By this stage in the subtopic, you should be getting the idea that reactions can go in both the forward and reverse directions and reach an equilibrium position when the rate of the forward reaction is equal to the rate of the reverse reaction. But what about a reaction that is not yet at equilibrium? Is there a way to determine in what direction a reaction not at equilibrium will proceed to reach its equilibrium position? By measuring the concentration of reactants and products at any time in a reaction, the reaction quotient, Q , can be calculated. Its value is then compared to the value of the equilibrium constant, K_c , and the direction in which the reaction will proceed can then be determined.

The reaction quotient is a measure of the relative amounts of reactants and products for a reaction that is not at equilibrium. It is used to determine in which direction a reaction will proceed to reach equilibrium and to compare to the equilibrium constant to see whether or not the system has reached equilibrium. The equation for the reaction quotient looks similar to that for the equilibrium constant in that it has the concentrations of the products in the numerator and the concentrations of the reactants in the denominator. Both concentrations of the reactants and products are raised to the powers of their coefficients in the balanced equation.

So for the general reaction between reactants A and B to form products C and D, where a , b , c and d are coefficients:

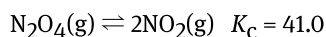


the equation for the reaction quotient is:

$$Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

The reaction quotient is calculated using the given concentrations of a system and is used to determine whether a reaction where the concentrations are changing is occurring, and in what direction that reaction is proceeding.

Take for example the reaction shown in which dinitrogen tetroxide, N_2O_4 , reacts to produce nitrogen dioxide, NO_2 .



The value of the equilibrium constant for this reaction indicates that the equilibrium position lies to the right and the equilibrium mixture contains a higher concentration of products than reactants. **Table 1** shows the concentrations of N_2O_4

and NO_2 in a new system, at a particular point in time. We can use these concentrations to calculate the value of the reaction quotient and determine in which direction the reaction will proceed to reach equilibrium.

Table 1. The concentrations of N_2O_4 and NO_2 at a particular time.

	$\text{N}_2\text{O}_4(\text{g})$	$\text{NO}_2(\text{g})$
Concentration at time t (mol dm^{-3})	2.00	6.00

Start by writing the expression for the reaction quotient for the reaction:

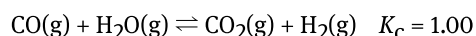
$$Q = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]}$$

Next, substitute in the concentrations of the reactant and products and calculate the value of the reaction quotient:

$$\begin{aligned} Q &= \frac{[6.00]^2}{[2.00]} \\ &= 18.0 \end{aligned}$$

Now that we have calculated the value of the reaction quotient, what does it tell us about the reaction? In this example the value of the reaction quotient is less than the value of the equilibrium constant. This means that the concentration of products is lower than at equilibrium, which means that the reaction will proceed to the right to reach equilibrium. This has the effect of increasing the concentrations of the products until the reaction reaches equilibrium. At this point the value of the reaction quotient is equal to the value of the equilibrium constant and the concentrations of reactants and products remain constant.

In the next example, we will look at the reaction between carbon monoxide and steam to produce carbon dioxide and hydrogen.



The concentrations of reactants and products at a particular time are shown in **Table 2**.

Table 2. The concentrations of reactants and products in the reaction between carbon monoxide and steam at a particular time.

	$\text{CO}(\text{g})$	$\text{H}_2\text{O}(\text{g})$	$\text{CO}_2(\text{g})$	$\text{H}_2(\text{g})$
Concentration at time t (mol dm^{-3})	0.500	0.500	1.00	1.00

First, write the equation for the reaction quotient:

$$Q = \frac{[\text{CO}_2][\text{H}_2]}{[\text{CO}][\text{H}_2\text{O}]}$$

Next, substitute in the concentrations of the reactant and products and calculate the value of the reaction quotient:

$$\begin{aligned} Q &= \frac{[1.00][1.00]}{[0.500][0.500]} \\ &= 4.00 \end{aligned}$$

In this example, the value of the reaction quotient is greater than the value of the equilibrium constant. This means that the reaction mixture has a higher concentration of products than the equilibrium mixture. Therefore, to reach equilibrium, the reaction will proceed to the left to increase the concentration of the reactants until the reaction reaches equilibrium.

The relationship between the reaction quotient and the equilibrium constant is summarised in **Figure 1**.

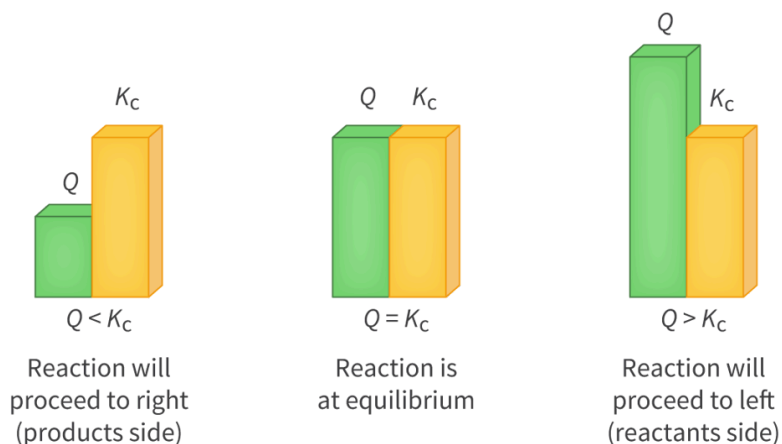


Figure 1. By comparing the value of the reaction quotient, Q , to the value of the equilibrium constant, K_c , for a reaction, the direction in which the reaction will proceed to reach equilibrium can be determined.

 More information for figure 1

The image shows a comparison of the reaction quotient (Q) and the equilibrium constant (K_c) through three bar graphs.

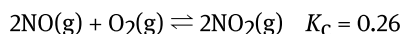
1. The leftmost graph features two bars, with the Q bar being shorter than the K_c bar. Below, it reads: ' $Q < K_c$. Reaction will proceed to right (products side).'
2. The center graph shows two bars of equal height for Q and K_c . The text below states: ' $Q = K_c$. Reaction is at equilibrium.'
3. The rightmost graph has a taller Q bar compared to the K_c bar. Below, it says: ' $Q > K_c$. Reaction will proceed to left (reactants side).'

[Generated by AI]

- When $Q > K_c$ the concentration of products is greater than at equilibrium. The reverse reaction is favoured and the reaction will proceed to the left to reach equilibrium.
- When $Q = K_c$ the reaction is at equilibrium. The rate of the forward reaction is equal to the rate of the reverse reaction.
- When $Q < K_c$ the concentration of reactants is greater than at equilibrium. The forward reaction is favoured and the reaction will proceed to the right to reach equilibrium.

Worked example 1

Nitrogen monoxide reacts with oxygen to form nitrogen dioxide according to the following equation:



At a particular point in time the concentrations of $\text{NO}(\text{g})$, $\text{O}_2(\text{g})$ and $\text{NO}_2(\text{g})$ are found to be 0.15 mol dm^{-3} , 0.80 mol dm^{-3} and $0.040 \text{ mol dm}^{-3}$ respectively.

Determine the value of the reaction quotient and predict in which direction the reaction will proceed to reach equilibrium.

First, write the equation for the reaction quotient:

$$Q = \frac{[\text{NO}_2]^2}{[\text{NO}]^2[\text{O}_2]}$$

Next, input the concentrations of reactants and products and calculate the value of Q .

$$\begin{aligned} Q &= \frac{[0.040]^2}{[0.15]^2[0.80]} \\ &= 0.0888 \dots \\ &= 0.089 \text{ (2 s.f.)} \end{aligned}$$

Finally compare the value of the reaction quotient with the value of the equilibrium constant.

In this example, $Q < K_{\text{c}}$ so the reaction will proceed to the right to reach equilibrium.

Use **Interactive 1** to test your understanding of the value of the reaction quotient, Q , and the direction of reaction.

	The direction of reaction to reach equilibrium	The change in the value of Q to reach equilibrium
$Q > K_{\text{c}}$		
$Q < K_{\text{c}}$		
$Q = K_{\text{c}}$		

The reaction will proceed to the left	The reaction is at equilibrium	The reaction will proceed to the right
Q will increase	Q will not change	Q will decrease

☒ Check

Interactive 1. The reaction quotient, Q , and the direction of reaction.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 25 minutes
- **Activity type:** Individual activity

Imagine you are attending a ballroom dancing event. The event is a little different because this event is for parents and guardians and their children. A dancing couple is made up of one adult and one child and, as the event goes on, different combinations of adults and children enter the dance floor and then leave the dance floor when they get tired and need to rest. The hall can accommodate a total of 60 adults and 60 children, but the dance floor is only large enough for 50 dancing couples, so the extra dancers must wait for a spot to open up.

1. If we think of this dance event as a system at equilibrium, identify what is required to make this a closed system.
2. Using the equation:



state the equilibrium expression for this system.

3. Deduce the value of the equilibrium constant, K_c , for this system at equilibrium and explain the significance of this value. Imagine that the hall has just opened the doors and the dancers have entered the hall, but the music has not yet begun, so there are no dancers yet on the dance floor. State the value of the reaction quotient, Q , for this scenario and determine to which side the reaction will proceed to reach equilibrium.
4. Describe a scenario where $Q > K_c$ for this analogy and explain how this affects the system.

1. No additional dancers may enter once the hall reaches its capacity of 120 people (60 adults and 60 children).
2. $K = [\text{couple}] / ([\text{child}][\text{adult}])$
3. At equilibrium, there will be 50 couples and the extra dancers waiting. Since a total of 60 adults and 60 children entered, there will be 10 adults and 10 children waiting ($60 - 50 = 10$)

$$K = \frac{50}{10 \times 10} = 0.5$$

The reverse reaction, couples leaving the dance floor, is favoured since $K < 1$.

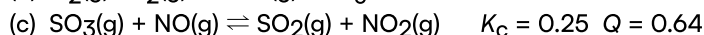
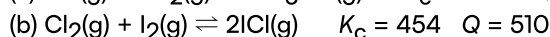
Adults and children will pair up to form a dancing couple until there are exactly 50 couples. The low K value indicates that once equilibrium is reached, the couples will leave the dance floor first and then be replaced by a new couple. Although the dance floor holds exactly 50 dancing couples at equilibrium, it takes more time for an adult and child to form a couple and

enter the dance floor than it does for a tired couple to leave the dance floor and separate.

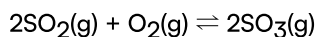
4. $Q = 0$, since there are no couples formed on the dance floor yet (no products). The reaction will proceed to the right/forward to reach equilibrium as adults and children pair up to form a couple.
5. If $Q > K$, then more than 50 couples are on the dance floor (greater concentration of products compared to reactants). This is greater than the number of dancing couples permitted, resulting in more couples leaving the dance floor than those entering. This would cause the reaction to shift to the left/reverse to reach equilibrium.

Now let's try some questions with chemical reactions:

1. For the following reactions, compare the values of Q and K_C and determine to which side the reaction will proceed to reach equilibrium.



2. Calculate the value of the reaction quotient, Q , for the following reaction.



	SO₂(g)
Concentration at time t (mol dm ⁻³)	0.254

The K_C for the reaction was found to be 4.62.

Determine in which direction the reaction will proceed to reach equilibrium.

1. (a) $Q < K_C$ so reaction will proceed to right
 (b) $Q > K_C$ so reaction will proceed to left
 (c) $Q > K_C$ so reaction will proceed to left

2.

$$\begin{aligned}
 Q &= \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]} \\
 &= \frac{[0.500]^2}{[0.254]^2[0.0855]} \\
 &= 45.321 \dots \\
 &= 45.3 \text{ (3 s.f.)}
 \end{aligned}$$

$Q > K_C$ so reaction will proceed to the left to reach equilibrium.

Calculations involving equilibrium reactions (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to solve problems involving values of the equilibrium constant, K_c , and initial and equilibrium concentrations of the components of an equilibrium mixture.

Have you ever tried walking up the down escalator in your local shopping centre? Let's hope not because not only is it against the rules but it is also very dangerous! But let's imagine you did try it. What would happen if you walked up at the same speed as the escalator is coming down? How would you get to the top?

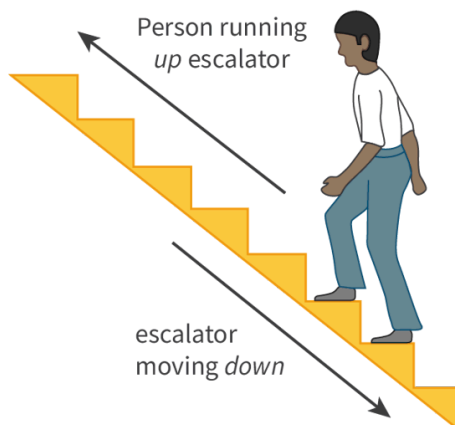


Figure 1. How is walking up an escalator that is moving down analogous to a reaction at equilibrium?

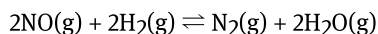
If you walked up the escalator at the same speed as the escalator was coming down, then you would stay in the same position, which is analogous to a chemical reaction at equilibrium. To get to the top, you would have to run at a faster speed than the escalator is coming down. This is analogous to the forward reaction happening faster than the reverse reaction.

In section R2.3.2–3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-equilibrium-constant-kc-and-magnitude-of-the-kc-id-46067/>) you learned that the equilibrium law allows the equilibrium constant, K_c , to be determined from the stoichiometry of the balanced equation for a reaction. In this section, you will learn how to calculate equilibrium concentrations of reactants and products of an equilibrium mixture.

Calculating equilibrium concentrations from initial concentrations

In the first example, you are given the initial concentrations of the reactants and products and the equilibrium concentration of at least one product. From this data, the equilibrium concentrations of the other species can be calculated, together with the equilibrium constant, K_c , for the reaction.

Nitrogen monoxide reacts with hydrogen to produce nitrogen and water according to the following equation:



0.100 mol of NO, 0.0500 mol of H_2 and 0.100 mol of H_2O are placed in a sealed 1.00 dm^3 flask at 25°C and allowed to reach equilibrium.

At equilibrium, 0.0620 mol of NO is present in the flask.

Determine the equilibrium concentrations of H_2 , N_2 and H_2O and calculate the value of the equilibrium constant, K_c .

Study skills

In these examples, you are given the amount (in mol) of reactants or products and the volume of the reaction vessel (usually 1.00 dm^3). To calculate the concentrations [section S1.4.5](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/>) of reactants and products, simply divide the amount (in mol) by the volume of the reaction vessel (in dm^3). Always check the volume of the reaction vessel in the question and do not assume that it is always going to be 1.00 dm^3 .

To calculate the equilibrium concentration of reactants and products we can make an ICE box. ICE stands for the initial concentration, change in concentration and equilibrium concentration, all in mol dm^{-3} . Once the ICE box is made, input the known concentrations. Note that if no value is given in the question, we assume the initial concentration is zero.

	2NO(g)	2H ₂ (g)	⇌
Initial concentration (mol dm^{-3})	0.100	0.0500	
Change in concentration (mol dm^{-3})			
Equilibrium concentration (mol dm^{-3})	0.0620		

Using the coefficients in the balanced equation, we can deduce that the change in the concentrations of H_2 and H_2O must be numerically the same as the change for the NO (0.0620 mol dm^{-3}). This is because the coefficients of the H_2 and H_2O are the same. The change in the concentration of the N_2 will be half this value because the coefficient is a one (which is half of two). Note, however, that the sign of the change

will be the opposite; the sign for the reactants will be negative because they are decreasing and the sign for the products are positive because they are increasing. The completed ICE box is shown below.

	2NO(g)	2H ₂ (g)	⇌
Initial concentration (mol dm ⁻³)	0.100	0.0500	
Change in concentration (mol dm ⁻³)	-0.0380	-0.0380	
Equilibrium concentration (mol dm ⁻³)	0.0620	0.0120	

Now that we have the equilibrium concentrations of the reactants and products we can calculate the value of the equilibrium constant.

First, write the equilibrium constant equation for the reaction:

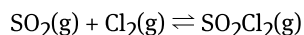
$$K_c = \frac{[\text{N}_2][\text{H}_2\text{O}]^2}{[\text{NO}]^2[\text{H}_2]^2}$$

Next, substitute in the concentrations of reactants and products and calculate the value of K_c .

$$\begin{aligned} K_c &= \frac{[0.0190][0.138]^2}{[0.0620]^2[0.0120]^2} \\ &= 653.681\dots \\ &= 654 \text{ (3 s.f.)} \end{aligned}$$

Worked example 1

0.80 mol of SO₂(g) and 1.60 mol of Cl₂(g) are placed in a 1.00 dm³ sealed flask and allowed to reach equilibrium.



At equilibrium 0.40 mol of SO₂Cl₂(g) are present.

Determine the equilibrium concentrations of SO₂(g) and Cl₂(g) and calculate the value of the equilibrium constant, K_c .

Construct an ICE box and determine the equilibrium concentrations of SO₂(g) and Cl₂(g).

	SO ₂ (g)	Cl ₂ (g)
Initial concentration (mol dm ⁻³)	0.80	1.60
Change in concentration (mol dm ⁻³)	-0.40	-0.40

	SO₂(g)	(
Equilibrium concentration (mol dm ⁻³)	0.40	

Write the equilibrium constant expression for the reaction:

$$K_c = \frac{[\text{SO}_2\text{Cl}_2]}{[\text{SO}_2][\text{Cl}_2]}$$

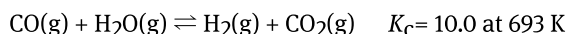
Next, substitute in the concentrations of reactants and products and calculate the value of K_c .

$$\begin{aligned} K_c &= \frac{[0.40]}{[0.40][1.20]} \\ &= 0.833\dots \\ &= 0.83 \text{ (2 s.f.)} \end{aligned}$$

Calculating equilibrium concentrations from initial concentrations and the equilibrium constant, K_c , for the reaction

In the next example, we will calculate equilibrium concentrations of reactants and products from initial concentrations and the equilibrium constant, K_c , for the reaction. The equilibrium concentrations of reactants and products can be determined by constructing an ICE box and solving for the unknown value, x .

Carbon monoxide reacts with steam according to the following equation:



1.00 mol of CO(g) and 1.00 mol of H₂O(g) are mixed in a 1.00 dm³ container. Calculate the concentration of each component in the equilibrium mixture.

First, construct an ICE box and input the initial concentrations of CO and H₂O. Because we do not know the equilibrium concentrations of any of the reactants or products, we will use x to represent the change. The change in concentration for CO(g) and H₂O(g) are both $1.00 - x$ and the change in concentration for H₂(g) and CO₂(g) are both $+x$.

	CO(g)	H₂O(g)	:
Initial concentration (mol dm ⁻³)	1.00	1.00	
Change in concentration (mol dm ⁻³)	-x	-x	
Equilibrium concentration (mol dm ⁻³)	1.00 - x	1.00 - x	

Next, substitute these values into the expression for the equilibrium constant and solve for x .

$$K_c = \frac{[\text{H}_2][\text{CO}_2]}{[\text{CO}][\text{H}_2\text{O}]}$$

$$= \frac{[x][x]}{[1.00 - x][1.00 - x]}$$

Substitute in the value of the equilibrium constant:

$$10.0 = \frac{x^2}{(1.00 - x)^2}$$

This is a quadratic equation, however there is a way to simplify this equation by taking the square root of both sides of the equation.

Take the square root of both sides of the equation:

$$\sqrt{10.0} = \sqrt{\frac{x^2}{(1.00 - x)^2}}$$

Solve for x :

$$\frac{x}{(1.0 - x)} = 3.16 \dots$$

$$x = 0.760$$

Finally, we use the value of x to determine the equilibrium concentrations of reactants and products.

$$[\text{CO}] = 1.00 - 0.760 = 0.240 \text{ mol dm}^{-3}$$

$$[\text{H}_2\text{O}] = 1.00 - 0.760 = 0.240 \text{ mol dm}^{-3}$$

$$[\text{H}_2] = x = 0.760 \text{ mol dm}^{-3}$$

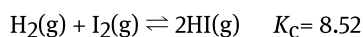
$$[\text{CO}_2] = x = 0.760 \text{ mol dm}^{-3}$$

Study skills

Note that the use of quadratic equations will not be assessed in IB DP chemistry.

Worked example 2

Calculate the equilibrium concentrations of $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ for this reaction if the initial concentrations are both 0.15 mol dm^{-3} .



Construct an ICE box for the reaction.

	$\text{H}_2(\text{g})$	I_2
Initial concentration (mol dm^{-3})	0.15	C
Change in concentration (mol dm^{-3})	$-x$	.

	H₂(g)	I₂
Equilibrium concentration (mol dm ⁻³)	0.15-x	0.1

Next, write the equilibrium constant expression:

$$\begin{aligned}
 K_c &= \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} \\
 &= \frac{[2x][2x]}{[0.15-x][0.15-x]} \\
 8.52 &= \frac{4x^2}{(0.15-x)^2}
 \end{aligned}$$

This is a quadratic equation, however there is a way to simplify this equation by taking the square root of both sides of the equation.:

$$\begin{aligned}
 \sqrt{8.52} &= \sqrt{\frac{4x^2}{(0.15-x)^2}} \\
 \frac{2x}{0.15-x} &= 2.92 \\
 x &= 0.089
 \end{aligned}$$

$$[\text{H}_2] = 0.15 - 0.089 = 0.061 \text{ mol dm}^{-3}$$

$$[\text{I}_2] = 0.15 - 0.089 = 0.061 \text{ mol dm}^{-3}$$

$$[\text{HI}] = 2(0.089) = 0.18 \text{ mol dm}^{-3}$$

Calculating equilibrium concentrations when the value of the equilibrium constant, K_c , is very small

In the final example we will calculate the equilibrium concentrations of reactants and products where the value of the *equilibrium constant* is very small. The small value means that the forward reaction barely proceeds so the concentration of products is very small. For this type of reaction, the approximation used is $[\text{reactant}]_{\text{initial}} \approx [\text{reactant}]_{\text{eqm}}$. This means that the equilibrium concentration of the reactants is approximately equal to the initial concentration.

Nature of Science

Aspect: Measurement

How valid are assumptions made in science? For example, the mass of an electron is often assumed to be negligible compared to the mass of a proton or a neutron. But how valid is this assumption? The mass of an electron is 9.109384×10^{-31} kg and the mass of a proton is 1.672622×10^{-27} kg. So a single proton has the same mass as approximately 1836 electrons. In this example, we can say the assumption is valid, which is why it is said that almost all of the mass of an atom is contained in its dense nucleus. Can you think of any other assumptions made in science? How valid are they?

Bromine and chlorine react to form bromine chloride as follows:



0.80 mol of $\text{Br}_2(\text{g})$ and 0.80 mol of $\text{Cl}_2(\text{g})$ are added to a 1.00 dm^3 reaction vessel and allowed to reach equilibrium. Calculate the equilibrium concentration of $\text{BrCl}(\text{g})$. Because the value of the K_c is very small, the assumption is $[\text{reactants}]_{\text{initial}} \approx [\text{reactants}]_{\text{eqm}}$. So the equilibrium concentrations of $\text{Br}_2(\text{g})$ and $\text{Cl}_2(\text{g})$ are assumed to be 0.80 mol dm^{-3} .

Start by constructing an ICE box and input the known values.

	$\text{Br}_2(\text{g})$	$\text{Cl}_2(\text{g})$	$\rightleftharpoons$	$2\text{BrCl}(\text{g})$
Initial concentration (mol dm^{-3})	0.80	0.80		
Change in concentration (mol dm^{-3})	$0.80 - x$	$0.80 - x$		+
Equilibrium concentration (mol dm^{-3})	≈ 0.80	≈ 0.80		;

Write the equilibrium constant equation:

$$K_c = \frac{[\text{BrCl}]^2}{[\text{Br}_2][\text{Cl}_2]}$$

Substitute in the known values and solve for x :

$$\begin{aligned} 8.00 \times 10^{-9} &= \frac{[2x]^2}{[0.80][0.80]} \\ 4x^2 &= 5.12 \times 10^{-9} \\ x &= 3.577 \times 10^{-5} \\ &= 3.6 \times 10^{-5} \\ &\text{(2s. f.)} \end{aligned}$$

The equilibrium concentration of $\text{BrCl}(\text{g})$ is $2x$ so:

$$[\text{BrCl}(\text{g})] = 2 \times 3.6 \times 10^{-5} = 7.2 \times 10^{-5} \text{ mol dm}^{-3}$$

The very low concentration of the product is logical because the equilibrium constant value was very small, meaning that the equilibrium position lies very much to the left.

If we make the ICE box again, but this time subtract x from the initial concentrations, we can see that the equilibrium concentration of Br_2 and Cl_2 are both 0.80 mol dm^{-3} when rounded to two significant figures. Therefore, the assumption that was made in the earlier calculation was valid.

	$\text{Br}_2(\text{g})$	$\text{Cl}_2(\text{g})$	$\rightleftharpoons$	$2\text{BrCl}(\text{g})$
Initial concentration (mol dm^{-3})	0.80	0.80		C

	$\text{Br}_2(\text{g})$	$\text{Cl}_2(\text{g})$	$\rightleftharpoons$	$2\text{BrCl}(\text{g})$
Change in concentration (mol dm^{-3})	$0.80 - \frac{3.6 \times 10^{-3}}{5}$	$0.80 - \frac{3.6 \times 10^{-3}}{5}$		$2 \times \frac{3.6 \times 10^{-3}}{5}$
Equilibrium concentration (mol dm^{-3})	0.80	0.80		7.2×10^{-3}

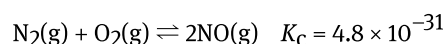
Study skills

The assumption $[\text{reactant}]_{\text{initial}} \approx [\text{reactant}]_{\text{eqm}}$ is used when the value of a very small number is added or subtracted from a larger value and there is no change in the value of the larger number when rounded to the appropriate number of significant figures. For example, when calculating the pH of a weak acid or base in [subtopic R3.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>).

Worked example 3

The reaction shown between nitrogen and oxygen is shown:



The initial concentrations of N_2 and O_2 are $0.033 \text{ mol dm}^{-3}$ and $0.0081 \text{ mol dm}^{-3}$ respectively.

Determine the equilibrium concentration of $\text{NO}(\text{g})$ and include any assumptions made in your calculations.

Construct an ICE box and input the initial concentrations of N_2 and O_2 .

Because the equilibrium constant is very small the assumption is $[\text{reactant}]_{\text{initial}} \approx [\text{reactant}]_{\text{eqm}}$.

	$\text{N}_2(\text{g})$	O_2
Initial concentration (mol dm^{-3})	0.033	0.0081
Change in concentration (mol dm^{-3})	$0.033 - x$	$0.0081 - x$
Equilibrium concentration (mol dm^{-3})	≈ 0.033	≈ 0.0081

Write the equilibrium constant equation for the reaction.

$$K_c = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$$

Substitute in the values and solve for x .

$$4.8 \times 10^{-31} = \frac{[2x]^2}{[0.033][0.0081]}$$

$$4x^2 = 1.28 \times 10^{-34}$$

$$x = 5.7 \times 10^{-18} \text{ (2 s.f.)}$$

$$\begin{aligned} [\text{NO(g)}] &= 2x \\ &= 2 \times 5.7 \times 10^{-18} \\ &= 1.1 \times 10^{-17} \text{ mol dm}^{-3} \end{aligned}$$

The assumption made in this calculation is that the equilibrium concentration is the same as the initial concentration.

The value of the equilibrium constant is very small meaning that the forward reaction has barely proceeded.

Making connections

How does the equilibrium law help us to determine the pH of a weak acid, weak base or a buffer solution?

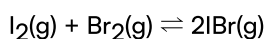
When calculating the pH of a weak acid, the assumption is that the equilibrium concentration of the weak acid is equal to the initial concentration of the weak acid. [Subtopic R3.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>)

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 25 minutes
- **Activity type:** Individual activity

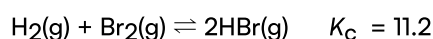
1. Iodine and bromine react to form iodine bromide according to the following equation.



0.40 mol of I_2 and 0.40 mol of Br_2 are mixed in a 1.00 dm^3 vessel. At equilibrium, 0.40 mol of IBr are present.

Calculate the value of the equilibrium constant for the reaction.

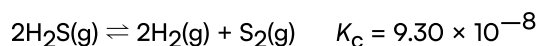
2. Hydrogen and bromine react to form hydrogen bromide according to the following equation.



0.600 mol of H_2 and 0.600 of Br_2 are mixed in a 1.00 dm^3 vessel.

Calculate the concentration of HBr at equilibrium.

3. Hydrogen sulfide gas, H_2S , reacts to produce hydrogen gas and disulfur gas according to the following equation.



0.15 mol of H_2S is placed in a 1.00 dm^3 reaction vessel and allowed to reach equilibrium.

Determine the equilibrium concentration of $\text{S}_2(\text{g})$.

State the assumption made in the calculation.

1.

	$\text{I}_2(\text{g})$
Initial concentration (mol dm^{-3})	0.40
Change in concentration (mol dm^{-3})	-0.20
Equilibrium concentration (mol dm^{-3})	0.20

$$\begin{aligned}
 K_c &= \frac{[\text{IBr}]^2}{[\text{I}_2][\text{Br}_2]} \\
 &= \frac{[0.40]^2}{[0.20][0.20]} \\
 &= 4.0
 \end{aligned}$$

2.

	$\text{H}_2(\text{g})$
Initial concentration (mol dm^{-3})	0.600
Change in concentration (mol dm^{-3})	-x
Equilibrium concentration (mol dm^{-3})	$0.600 - x$

$$\begin{aligned}
 K_c &= \frac{[\text{HBr}]^2}{[\text{H}_2][\text{Br}_2]} \\
 &= \frac{[2x]^2}{[0.600 - x][0.600 - x]} \\
 \frac{(2x)^2}{(0.600 - x)^2} &= 11.2 \\
 \sqrt{\frac{(2x)^2}{(0.600 - x)^2}} &= \sqrt{11.2} \\
 \frac{(2x)}{(0.600 - x)} &= 3.346 \\
 x &= 0.376
 \end{aligned}$$

$$[\text{HBr}] = 2x = 0.752 \text{ mol dm}^{-3}$$

3. For this reaction the value of the equilibrium constant is very small, so the assumption is $[\text{reactant}]_{\text{initial}} \approx [\text{reactant}]_{\text{eqm}}$.

	$2\text{H}_2\text{S}(\text{g})$
Initial concentration (mol dm^{-3})	0.15
Change in concentration (mol dm^{-3})	$-2x$
Equilibrium concentration (mol dm^{-3})	≈ 0.15

$$\begin{aligned}
 K_c &= \frac{[\text{H}_2]^2[\text{S}_2]}{[\text{H}_2\text{S}]^2} \\
 9.30 \times 10^{-8} &= \frac{[2x]^2[x]}{[0.15]^2} \\
 4x^3 &= 2.09 \times 10^{-9} \\
 x &= 8.06 \times 10^{-4}
 \end{aligned}$$

$$[\text{S}_2] = 8.06 \times 10^{-4} \text{ mol dm}^{-3}$$

Equilibrium and Gibbs energy (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to calculate the value of the Gibbs energy change ΔG from the equilibrium constant for a reaction.

If a rock is rolled down a hill, where does it stop? The rock will stop rolling when it gets to the lowest point which is the lowest energy position as shown in **Figure 1**. To continue rolling up the other side would require an input of energy as would returning to its original position. A rock rolling down a hill is comparable to a spontaneous reaction [section R1.4.2–3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/) which is a reaction that proceeds without the addition of energy once the initial energy barrier has been overcome.

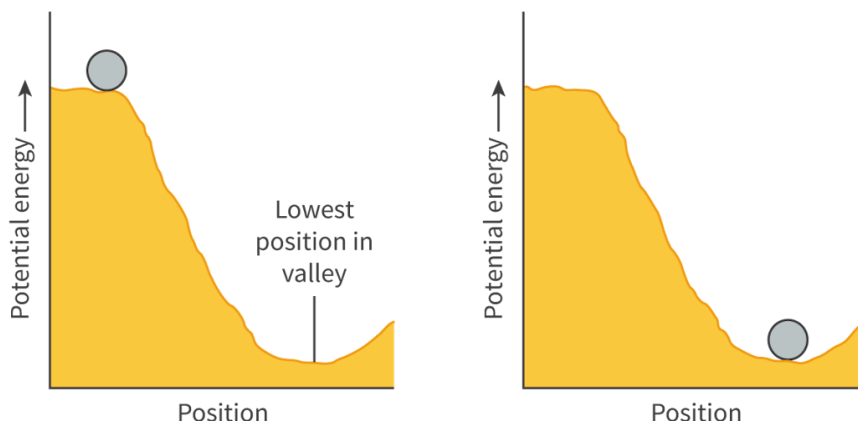


Figure 1. A rock rolling down a hill reaches the lowest energy point.

This is analogous to a chemical reaction proceeding towards equilibrium.

 More information for figure 1

The image consists of two graphical diagrams side by side, both illustrating potential energy in relation to position. In the left diagram, a rock is positioned at the top of a hill with high potential energy and begins to roll down a slope. The slope dips into a valley labeled as the "Lowest position in valley," where the potential energy is at its minimum. The right diagram shows the rock at rest in this valley, reaching equilibrium at the lowest energy point, illustrating a stable position with low potential energy. Both diagrams use an upward triangular shape to indicate the change in potential energy and demonstrate a physical analogy to a chemical reaction reaching a state of equilibrium.

[Generated by AI]

Figure 2 and **Figure 3** shows the change in Gibbs energy (see section R1.4.2–3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/>)) for a spontaneous and a non-spontaneous reaction, respectively. For both reactions the Gibbs energy reaches a minimum at which point the reaction is at equilibrium and the Gibbs energy change, ΔG , is zero. However, one important difference is the point in the reaction where the minimum position of Gibbs energy lies.

Study skills

A reaction at equilibrium has a minimum value of Gibbs energy and a maximum value of entropy.

For a spontaneous reaction (**Figure 2**), the minimum value of Gibbs energy lies further towards the products side meaning that the equilibrium mixture will contain mostly products. The equilibrium constant for a spontaneous reaction is greater than one, with its value increasing as a reaction becomes more spontaneous.

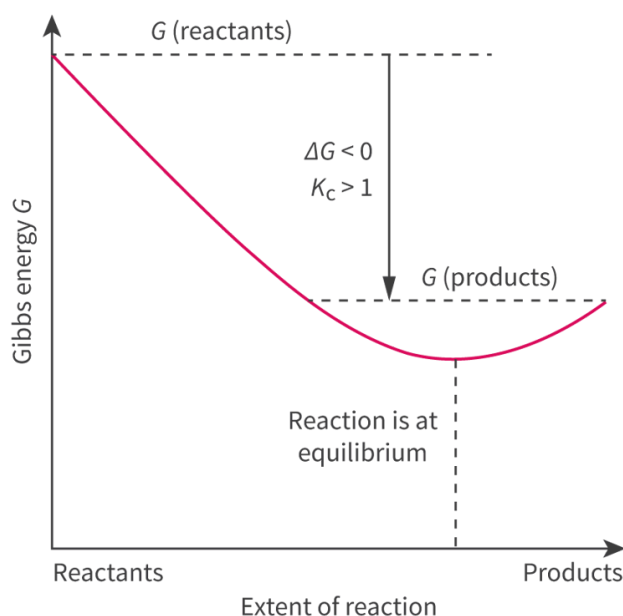


Figure 2. For a spontaneous reaction, the minimum point of Gibbs energy lies towards the products side.

 More information for figure 2

The image is a graph depicting the Gibbs energy (G) as a function of the extent of reaction. The X-axis is labeled 'Extent of reaction' and is divided into 'Reactants' on the left and 'Products' on the right. The Y-axis is labeled 'Gibbs energy G '. The graph shows a curve that starts high at the reactants side, indicating an initial high Gibbs energy. It slopes downward and reaches a minimum point towards the products side, indicating a lower Gibbs energy there. This signifies that the reaction is spontaneous. The energy difference is marked as $\Delta G < 0$. Text in the graph indicates " $K_c > 1$ " and "Reaction is at equilibrium" at the lowest point of the curve, which is closer to the products side than the reactants.

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For a non-spontaneous reaction (**Figure 3**), the minimum value of the Gibbs energy lies further towards the reactants side meaning that the equilibrium mixture will contain mostly reactants. The equilibrium constant for a non-spontaneous reaction is less than one, with its value decreasing as a reaction becomes less spontaneous.

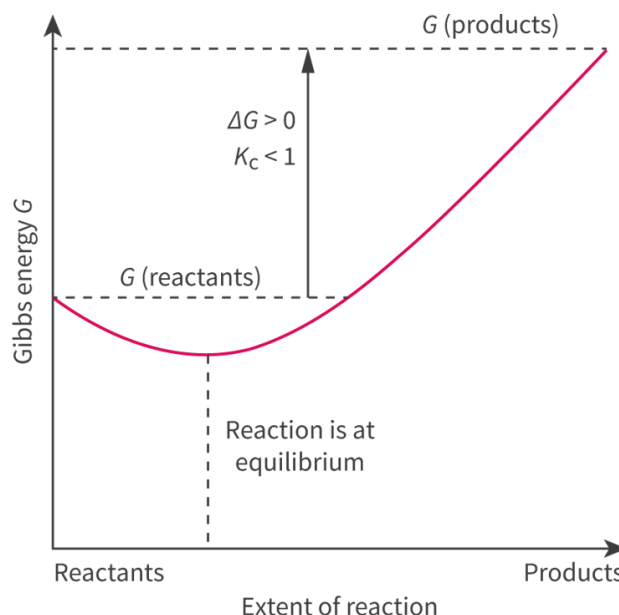


Figure 3. For a non-spontaneous reaction, the minimum point of Gibbs energy lies towards the reactants side.

 More information for figure 3

The graph represents the relationship between Gibbs energy (G) and the extent of a reaction for a non-spontaneous process. The X-axis is labeled as "Extent of reaction" and ranges from "Reactants" to "Products." The Y-axis is labeled as "Gibbs energy G ."

The graph features a curve that starts from the left at a higher energy level, decreases to a minimum point, and then rises again towards the right. The minimum point of the Gibbs energy curve lies towards the reactants side, indicating the position of equilibrium.

Annotations on the graph include a vertical dashed line at the minimum point labeled "Reaction is at equilibrium," showing where the equilibrium position is. The labels " G (reactants)" and " G (products)" are positioned horizontally on the left and right sides, respectively.

Additional annotations include $\Delta G > 0$ and $K_c < 1$, indicating that the reaction is non-spontaneous and the equilibrium constant (K_c) is less than one, consistent with a reaction where the products are not favored.

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The relationship between the ΔG , the equilibrium constant K_c and the composition of the equilibrium mixture is summarised in **Table 1**.

Table 1. The relationship between the ΔG , the equilibrium constant K_c and the composition of the equilibrium mixture.

$\Delta G^\ominus$	K_c	Composition of reaction mixture
< 0	> 1	Higher concentration of products
> 0	< 1	Higher concentration of reactants
0	1	Neither reactants nor products favoured

Making connections

How can Gibbs energy be used to explain whether the forward or backward reaction is favoured before reaching equilibrium (see subtopic R1.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-hl-id-43477/>))?

Calculating ΔG from the equilibrium constant

Recall that the Gibbs energy change, ΔG , can be used to determine whether a reaction will occur spontaneously at a particular temperature (see section R1.4.2–3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/>)). In this section you will learn about the relationship between the Gibbs energy change, ΔG , and the equilibrium position. The Gibbs energy change can be calculated from the value of the equilibrium constant, K_c , for a reaction using the following equation, which can be found in section 1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>) of the DP chemistry data booklet.

Table 2. Equation for standard Gibbs energy change.

Equation	Symbols	Units
$\Delta G^\ominus = -RT \ln K$	$\Delta G^\ominus$ is the standard Gibbs energy change	joules (J)
	R is the universal gas constant ($8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)	joules per kelvin mole ($\text{J K}^{-1} \text{ mol}^{-1}$)
	T is the temperature	kelvin (K)
	$\ln K$ is the natural log of the equilibrium constant, K_c	no unit

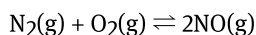
In this equation $\Delta G^\ominus$ has units of J mol^{-1} , to convert to kJ mol^{-1} , divide by 1000.

The equation can be rearranged to calculate the value of the equilibrium constant, K_c , if the Gibbs energy change is known:

$$K = e^{-\frac{\Delta G}{RT}}$$

Next, we will look at some example calculations, starting with calculating the Gibbs energy change, ΔG , for a reaction using the value of the equilibrium constant.

Nitrogen and oxygen react together to form nitrogen monoxide as shown.



The value of the equilibrium constant for this reaction at 150 °C is 1.60×10^{-3} .

Determine the value of the Gibbs energy change, ΔG , in kJ mol^{-1} .

$$\Delta G^\ominus = -RT \ln K$$

Substitute the values into the equation, making sure that the temperature is in kelvin (K).

$$\begin{aligned}\Delta G &= -8.31 \times (150.0 + 273.15) \times (\ln 1.60 \times 10^{-3}) \\ &= 22\,630 \text{ J mol}^{-1}\end{aligned}$$

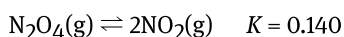
Convert to kJ mol^{-1} by dividing by 1000:

$$\Delta G = 22.6 \text{ kJ mol}^{-1} \text{ (to three significant figures)}$$

The positive ΔG value means the reaction is non-spontaneous and the equilibrium position lies to the left which is shown by the small value of the equilibrium constant.

Worked example 1

Calculate the value of the Gibbs energy change, ΔG , in kJ mol^{-1} , for this reaction at 298 K.



$$\Delta G^\ominus = -RT \ln K$$

$$\begin{aligned}\Delta G^\ominus &= -8.31 \times 298 \times \ln 0.140 \\ &= 4.87 \times 10^3 \text{ J mol}^{-1} \\ &= 4.87 \text{ kJ mol}^{-1}\end{aligned}$$

Next we will look at how to calculate the value of the equilibrium constant for a reaction using the value of the Gibbs energy change, ΔG .

The Haber process has a Gibbs energy change, ΔG , of $-33.0 \text{ kJ mol}^{-1}$ at 25.0 °C.

Calculate the value of the equilibrium constant at this temperature.

$$K = e^{-\frac{\Delta G}{RT}}$$

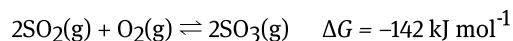
Substitute in the values and calculate the value of K . Note the value of the ΔG is converted to J mol^{-1} by multiplying by 1000 and the temperature is converted to kelvin by adding 273.15.

$$\begin{aligned}K &= e^{-\frac{-33.0 \times 1000}{8.31 \times (25.0 + 273.15)}} \\ &= e^{-(-13.3)} \\ &= 5.97 \times 10^5\end{aligned}$$

This value of K is expected because the ΔG is negative meaning that the equilibrium position lies to the right and the equilibrium mixture contains a higher concentration of products.

Worked example 2

Calculate the value of the equilibrium constant for this reaction at 298 K.



$$K = e^{-\frac{\Delta G}{RT}}$$

Don't forget to convert the ΔG to J mol^{-1} . The temperature is already given in Kelvin.

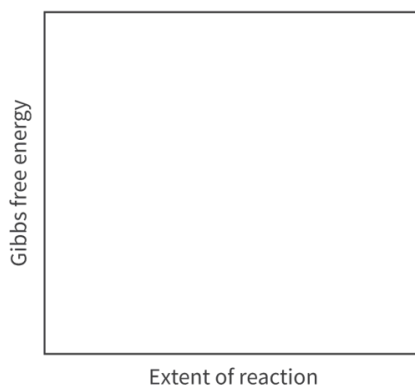
$$\begin{aligned} K &= e^{-\frac{-142 \times 1000}{8.31 \times 298}} \\ &= e^{-(-57.3)} \\ &= 7.67 \times 10^{24} \end{aligned}$$

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

1. On the axis provided below, sketch the change in Gibbs energy for a spontaneous and a non-spontaneous reaction. On each graph, label the equilibrium position.

Spontaneous reaction



 More information

The image contains a graph with the X-axis labeled "Extent of reaction" and the Y-axis labeled "Gibbs free energy". This graph is intended to show the change in Gibbs energy for both a spontaneous and a non-spontaneous reaction.

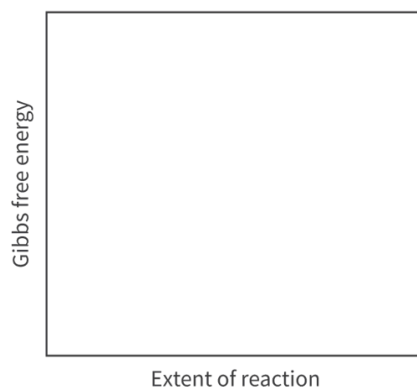
The task is to sketch:

- A spontaneous reaction: Typically, the curve would show a decrease in Gibbs free energy as the extent of the reaction progresses, indicating that the reaction releases energy and moves towards equilibrium.
- A non-spontaneous reaction: The curve would likely show an increase or no significant change in Gibbs free energy, suggesting that the reaction requires added energy to proceed and does not naturally progress towards equilibrium.

The equilibrium position on each graph should be clearly labeled, marking the point where the reaction is balanced with no net change in Gibbs free energy.

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Non-spontaneous reaction



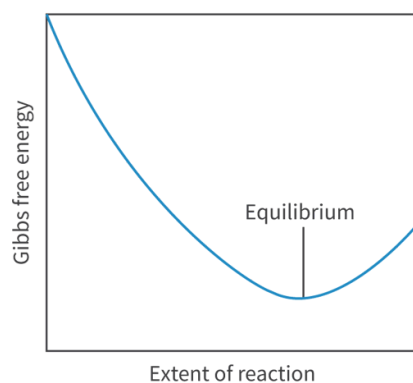
 More information

The image is a graph plotting 'Gibbs free energy' (y-axis) against 'extent of reaction' (x-axis). The graph represents how the Gibbs free energy changes as a reaction progresses. The axes are labeled with pertinent scientific terms essential for interpreting chemical thermodynamics. However, no specific data points, curves, or numerical values are visibly marked on the graph.

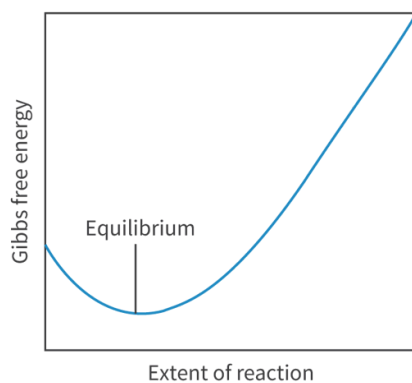
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- Calculate the Gibbs energy change, ΔG , in kJ mol^{-1} , for a reaction that has an equilibrium constant of 54.3 at 430°C .
- Calculate the equilibrium constant for a reaction that has a Gibbs energy change, ΔG , of 41.8 kJ mol^{-1} at 25.0°C .

1. Spontaneous reaction



Non-spontaneous reaction



$$2. \Delta G = -RT \ln K$$

$$\begin{aligned} \Delta G &= -8.31 \times (431 + 273.15) \times (\ln 54.3) \\ &= -23369 \text{ J mol}^{-1} \end{aligned}$$

Convert to kJ mol^{-1} by dividing by 1000:

$$\Delta G = -23.4 \text{ kJ mol}^{-1}$$

3.

$$K_c = e^{-\frac{\Delta G}{RT}}$$

$$\begin{aligned} K_c &= e^{-\frac{11.8 \times 1000}{8.31 \times (25.0 + 273.15)}} \\ &= e^{-(16.9)} \\ &= 4.58 \times 10^{-8} \end{aligned}$$